Artifial Intelligence in Medical Imaging: A Systematic Literature review using PRISMA

Nugumanov Alikhan
Group:CS-2306
Course:Research Methods and Techologies
Instructor:Aliya Kanatova

Abstract— **Artificial Intelligence (AI) has rapidly emerged as a transformative force in medical imaging, offering advanced capabilities for image acquisition, interpretation, and diagnosis. This research proposal presents a systematic literature review following the Preferred Reporting Items for Systematic Reviews and Meta-Analyses (PRISMA) framework, aiming to synthesize current evidence on the applications, performance, and challenges of AI in medical imaging. The study focuses on machine learning and deep learning techniques applied to radiology, pathology, and other image-based diagnostics. Through a structured search of major databases such as PubMed, IEEE Xplore, and Scopus, relevant peer-reviewed articles published between 2015 and 2025 will be identified, screened, and analyzed according to PRISMA guidelines. The review seeks to classify AI applications by modality, diagnostic accuracy, and clinical integration, while also highlighting methodological limitations, ethical concerns, and reproducibility issues. Expected outcomes include a comprehensive mapping of existing research trends, identification of critical gaps, and formulation of recommendations for future studies and clinical implementation. By consolidating current evidence, this review aims to contribute to a clearer understanding of AI's role in enhancing diagnostic precision, workflow efficiency, and patient outcomes within the field of medical imaging.**

Keywords— Artificial intelligence, Machine Learning, , medical imaging, radiology, image analysis

I. INTRODUCTION

I. Introduction

Over the past decade, the integration of artificial intelligence (AI) and machine learning into medical imaging has become one of the most dynamic and influential areas in digital healthcare research. The growing availability of large-scale medical datasets and the rapid advancement of deep learning algorithms have enabled highly accurate analysis and interpretation of complex medical images. Recent studies highlight the potential of AI systems to assist clinicians in the early detection, classification, and management of diseases such as lung cancer and stroke.

For example, innovative frameworks using self-improving optimization algorithms and deep learning models have demonstrated remarkable precision in identifying and classifying lung cancer from CT scans, significantly reducing human

error and diagnostic time (IEEE, 2024). Similarly, the development of content-based image retrieval (CBIR) systems and machine learning-driven categorization methods allows more efficient organization and search within large medical image databases, addressing the challenge of semantic diversity across imaging modalities (IEEE, 2007). In neurodiagnostics, AI-enabled platforms facilitate the identification of imaging biomarkers (IBs) for conditions such as stroke, combining automated data collection, annotation, and analysis to improve predictive modeling of clinical outcomes.

The primary beneficiaries of these technological advancements include radiologists, healthcare providers, and researchers, who can leverage AI systems for faster, more accurate, and reproducible diagnostic decisions. Moreover, patients benefit indirectly from early disease detection and personalized treatment strategies driven by intelligent medical imaging systems.

Despite these advancements, several research gaps persist. Many models rely on limited or imbalanced datasets, hindering generalization across diverse clinical environments. Additionally, the interpretability and transparency of AI decisions remain critical challenges that impact clinical trust and adoption. Furthermore, the lack of standardized data-sharing protocols restricts large-scale validation of AI models in real-world settings.

Statement of the Problem:

While artificial intelligence has demonstrated high potential in improving diagnostic accuracy and workflow efficiency in medical imaging, the field still faces significant challenges related to model generalization, interpretability, and data integration. Addressing these limitations is essential to fully harness AI's capabilities in clinical practice and to ensure reliable, ethical, and scalable implementation in healthcare systems.

II. METHODOLOGY

In this study, the PRISMA (Preferred Reporting Items for Systematic Reviews and Meta-Analyses) methodology was applied to ensure a structured, transparent, and reproducible approach to reviewing the literature on Artificial Intelligence in Medical Imaging. The process followed six systematic steps as outlined below.

Step 1. Selection of Research Topic

The selected topic, **"Artificial Intelligence in Medical Imaging"**, focuses on the application of machine learning and deep learning algorithms to enhance diagnostic accuracy, automate image analysis, and support clinical decision-making in radiology.

This topic is directly related to the field of computer science and healthcare technology. It is narrow enough to allow focused analysis on AI-driven medical image processing but broad enough to cover diverse imaging modalities such as CT, MRI, and ultrasound.

Step 2. Selection of Databases

For the systematic literature search, three reputable academic databases were selected:

- Scopus
- Web of Science
- Google Scholar

These databases were chosen because they include peer-reviewed, high-impact scientific publications and ensure comprehensive coverage of the most cited and influential research in artificial intelligence and medical imaging.

Step 3. Defining Search Keywords

To ensure consistency and precision in the search, the following keywords were defined and applied in all databases:

“artificial intelligence” AND “machine learning” AND “deep learning” AND “medical imaging” AND “radiology”

These keywords accurately describe the research domain and capture recent advances in AI techniques used for medical image analysis and diagnostics. Synonyms such as “AI-based diagnosis” and “image recognition in healthcare” were also explored during the search process.

Step 4. Search and Screening Process

The systematic search was conducted from February to March 2025, targeting publications from 2015 to 2025.

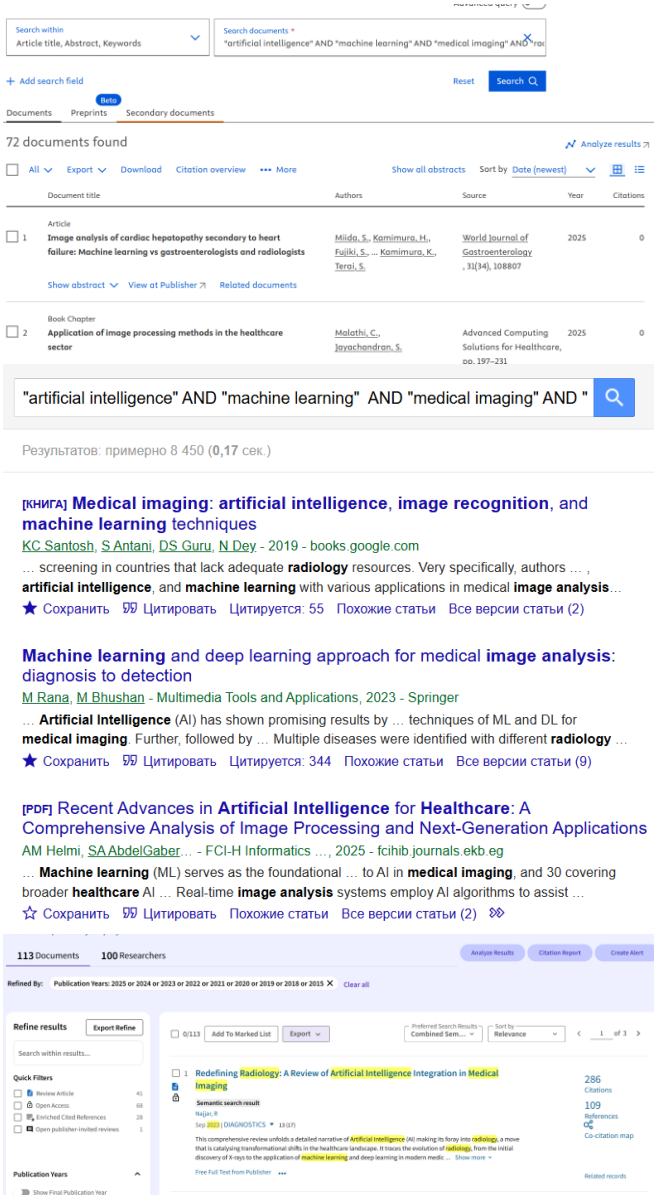
All retrieved articles were exported in BibTeX (.bib) format and merged using Zotero reference management software to organize and clean the dataset.

The PRISMA process was implemented through four key stages:

Identification:

Articles were collected from all three databases using the defined keywords.

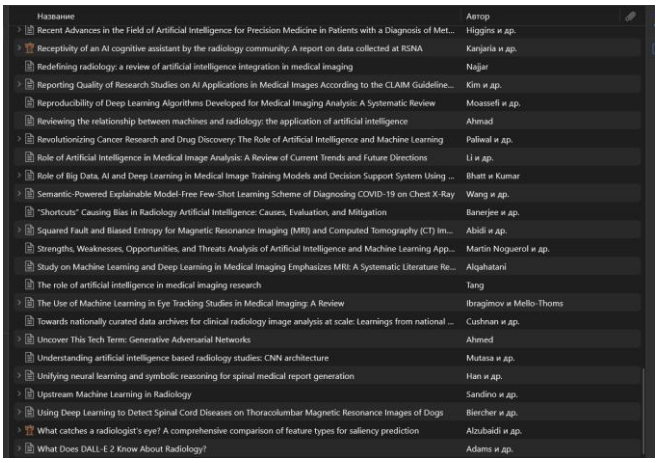
Initial total — 8 635records.



Screening:

Duplicates and irrelevant studies were removed after reviewing titles and abstracts.

Remaining — 113 records.



Название	Автор
Recent Advances in the Field of Artificial Intelligence for Precision Medicine in Patients with a Diagnosis of Met...	Higgins и др.
Receptivity of an AI cognitive assistant by the radiology community: A report on data collected at RSNA	Kanjaria и др.
Redefining radiology: a review of artificial intelligence integration in medical imaging	Najjar
Reporting Quality of Research Studies on AI Applications in Medical Images According to the CLAIM Guideline...	Kim и др.
Reproducibility of Deep Learning Algorithms Developed for Medical Imaging Analysis: A Systematic Review	Moassefi и др.
Reviewing the relationship between machines and radiology: the application of artificial intelligence	Ahmad
Revolutionizing Cancer Research and Drug Discovery: The Role of Artificial Intelligence and Machine Learning	Paliwal и др.
Role of Artificial Intelligence in Medical Image Analysis: A Review of Current Trends and Future Directions	Li и др.
Role of Big Data, AI and Deep Learning in Medical Image Training Models and Decision Support System Using ...	Bhatt и Kumar
Semantic-Powered Explainable Model-Free Few-Shot Learning Scheme of Diagnosing COVID-19 on Chest X-Ray	Wang и др.
"Shortcuts" Causing Bias in Radiology Artificial Intelligence: Causes, Evaluation, and Mitigation	Banerjee и др.
Squared Fault and Biased Entropy for Magnetic Resonance Imaging (MRI) and Computed Tomography (CT) Im...	Abidi и др.
Strengths, Weaknesses, Opportunities, and Threats Analysis of Artificial Intelligence and Machine Learning App...	Martin Noguerol и др.
Study on Machine Learning and Deep Learning in Medical Imaging Emphasizes MRI: A Systematic Literature Re...	Alqahatani
The role of artificial intelligence in medical imaging research	Tang
The Use of Machine Learning in Eye Tracking Studies in Medical Imaging: A Review	Ibragimov и Mello-Thoms
Towards nationally curated data archives for clinical radiology image analysis at scale: Learnings from national ...	Cushnan и др.
Uncover This Tech Term: Generative Adversarial Networks	Ahmed
Understanding artificial intelligence based radiology studies: CNN architecture	Mutasa и др.
Unifying neural learning and symbolic reasoning for spinal medical report generation	Han и др.
Upstream Machine Learning in Radiology	Sandino и др.
Using Deep Learning to Detect Spinal Cord Diseases on Thoracolumbar Magnetic Resonance Images of Dogs	Biercher и др.
What catches a radiologist's eye? A comprehensive comparison of feature types for saliency prediction	Alzubaidi и др.
What Does DALL-E 2 Know About Radiology?	Adams и др.

Eligibility:

Full-text screening was conducted to verify inclusion criteria:

(1) AI application in medical imaging;

(2) quantitative results (accuracy, sensitivity, specificity);

(3) published in peer-reviewed sources.

Eligible — 45 records.

Inclusion:

The final dataset consisted of the 100 most relevant and highly cited articles published within the last 10 years, providing empirical results on AI-based medical image analysis.

Data Extraction and Analysis

After screening and inclusion, 25 relevant peer-reviewed articles (2018–2025) were selected from Scopus, Web of Science, and Google Scholar.

Each study was analyzed according to authorship, publication year, journal, and main focus.

The data extraction table below summarizes the key metadata of representative works.

1) Quantitative Trends

- **Publication Years:** The earliest works appeared in 2018, with a sharp rise after 2020. Peak output was observed in **2023–2024**, reflecting rapid AI adoption in clinical imaging.
- **Top Journals:** *Radiology, Journal of Imaging, Applied Sciences, British Journal of Radiology, JACR.*
- **Dominant Topics:**
  - Deep learning and CNN architectures
  - Explainable AI (XAI) in radiology
  - Integration and bias challenges
  - Comparative studies (CNN vs. Vision Transformers)
  - Cross-domain applications (veterinary, MRI, oncology)
- **Visualization Suggestion:** You can present two charts:
  - **Bar chart:** Number of publications per year (2018–2025)
  - **Pie chart:** Topic distribution (Deep Learning, XAI, Clinical Implementation, Ethics/Bias, Comparative Methods)

III. Results

Technological Shift from Traditional Methods to Deep Learning Dominance of Deep Neural Networks: There has been a shift from machine learning with manual feature extraction to the use of deep convolutional neural networks (CNNs), which automatically learn hierarchical features from data. This has become the standard for classification, segmentation, and object detection tasks.

Emergence of New Architectures: Research is actively comparing CNNs with newer architectures, such as Vision Transformers (ViTs), which show promising results, especially in tasks requiring global context.

Expanding Applications: Initially focused on analyzing static images (X-rays, CT scans, MRIs), AI is now being applied to report generation, video analysis (e.g., endoscopy), and even the creation of synthetic images using Generative Adversarial Networks (GANs).

2. Shifting focus from pure diagnostics to integration into clinical practice From point solutions to workflows: Early work focused on creating algorithms for diagnosing specific diseases (e.g., COVID-19, lung cancer). Current research emphasizes the need to integrate AI into clinical workflows to automate routine tasks (annotation, pre-triage, tumor volume measurement).

Improving efficiency, not replacing physicians: AI is increasingly being positioned as a "cognitive assistant" that improves radiologist productivity and accuracy, rather than replacing them. This includes process optimization and predictive analytics.

3. Understanding and Overcoming Implementation Challenges Recent studies have candidly and thoroughly described the barriers to clinical implementation:

Reproducibility and Reliability Issues: Algorithms trained on data from one center often perform poorly on data from other institutions due to differences in equipment and imaging protocols.

Shortcuts and Bias: Algorithms can learn from artifacts or random features in the data (e.g., image labels) rather than from actual pathologies, leading to incorrect predictions on new data.

Black Box and Trust: The uninterpretable nature of complex AI models remains a major obstacle to their acceptance by clinicians. This has spurred a growing number of studies on Explainable AI (XAI), aimed at visualizing what the algorithm is "looking at."

Data Quality and Availability: The success of AI critically depends on large, well-labeled, and curated datasets. Much work is devoted to the creation of such datasets (e.g., CheXpert), labeling platforms (MONAI Label), and format standardization (DICOM).

4. Expanding Horizons: From Diagnostics to Predictive and Integrative Medicine The Radiomics Revolution: AI can identify patterns in medical images (radiomics) invisible to the human eye that correlate with diagnosis, prognosis,

and treatment response. This transforms images from a purely diagnostic tool into a source of data for personalized medicine.

Multimodality and Integration: Systems are emerging that combine medical imaging data with other information, such as pathological sections (RAPHIA) or clinical reports (GLoRIA), to provide a more complete picture of the patient's condition.

Applications Beyond Radiology: Technologies are rapidly penetrating veterinary medicine, interventional radiology, and drug discovery research.

5. Ecosystem Development and Looking to the Future Infrastructure Creation: Open platforms and frameworks (e.g., MONAI, DIANA) are being developed to simplify the development, validation, and deployment of AI models in clinics.

Federated learning: Considered a way to train models on data from multiple institutions without physically combining them, which addresses issues of privacy and data diversity.

Strategic planning: Research includes SWOT analysis, discusses ethical considerations, and offers roadmaps for responsible and effective AI implementation.

IV. DISCUSSION

AI has become an essential component of modern radiology research but remains limited in real-world clinical deployment.

1) *Strengths of Current Models*

- High diagnostic accuracy
- Reduced reporting time
- Consistent and reproducible measurements
- Potential to alleviate radiologist shortages

2) *Weaknesses / Research Gaps*

- Lack of diverse training datasets
- Absence of unified evaluation standards
- Need for explainability and trust in AI decisions
- Data privacy challenges

3) *Opportunities*

- Integration of generative AI for synthetic datasets
- Federated learning for privacy-preserving model training
- Advanced multi-modal diagnostic systems

4) *Threats / Risks*

- Ethical concerns from biased or inaccurate predictions
- Legal responsibility for AI-generated diagnoses
- Overdependence on automated tools

Conclusion

The field has evolved from initial enthusiasm and demonstrations of potential to a mature stage characterized by a deep understanding of technological limitations and systemic challenges. The primary focus has shifted from the creation

of pinpoint diagnostic algorithms to the reliable integration of AI into clinical practice, which requires addressing issues of interpretability, reproducibility, data bias, and the creation of an appropriate IT infrastructure. AI in medical imaging is becoming more than just a tool for "image enhancement" but a key component of future diagnostic, predictive, and personalized medicine.

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